



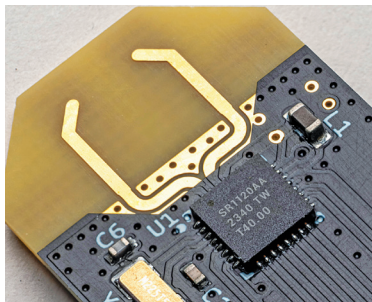
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Ultra wideband transceiver

SPARK Microsystems' SR1120 ultra wideband (UWB) transceiver addresses a challenge in wireless design by delivering low-power, low-latency data



with robust interference resistance. The transceiver doubles data throughput versus Bluetooth to 40.96Mbps, 20 to 40 times faster, while using about 25 times less power and reducing latency by 60 times, and it keeps the advantages over Wi-Fi. It also supports ranging at about 100 times lower power than other UWB solutions, all within a cost-efficient design. Compared with its predecessor, the SR1120 extends range by about 50 per cent. These changes allow new uses in gaming, edge AI, audio, and IoT applications, including industrial systems, robotics, and medical devices, and prepare body area networks to handle the data needs of implantable sensors and real-time neural activity transmission.

SPARK Microsystems
<https://www.sparkmicro.com>

System-on-chip

Ambiq Micro's Atomiq is a system-on-chip that lets devices run AI continuously at the edge without draining battery or adding heat. It uses sub- and near-threshold voltages to keep power low while still handling demanding tasks through its ARM Ethos-U85 NPU, which supports on-device computer

vision, speech recognition, and sensor processing. Dynamic power scaling adapts energy use to workload, making



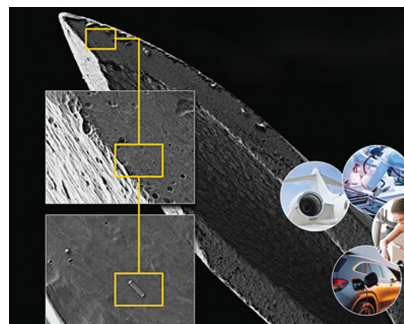
always-on AI practical for small, battery-powered devices. This approach opens possibilities for smart cameras and security systems that recognise objects in

real time, AR glasses and wearables with voice interfaces, and industrial or robotic systems that analyse data locally instead of relying on the cloud.

Ambiq
<https://ambiq.com>

Nanoscale temperature and force sensors

Sensors from Digid measure temperature and force at scales down to 1µm, with the company targeting future versions at 10nm, for use in areas such



as physical AI and humanoid robotics, where tactile and thermal feedback limits machine interaction. The sensors can be printed as arrays directly onto surfaces, allowing machines to detect touch and temperature closer to human sensing, and the same

devices can be used for force sensing on surgical scalpel blades, temperature measurement at the tips of probes, thermal monitoring inside battery cells, and biosensing tasks such as detecting viruses or chemical markers in the bloodstream. Digid says the sensors produce little self-heating, provide linear measurements, and remove the need for drift compensation and digital overhead found in conventional sensors.

Digid
<https://www.digid.com>

AI vision system-on-chip

The CV7 edge AI vision system-on-chip (SoC) from Ambarella, Inc. is designed for AI perception workloads across consumer, enterprise, industrial,



and automotive systems, supporting use cases such as 8K action and 360-degree cameras, multi-imager security cameras, drones, robotics, industrial automation, video conferencing, and automotive platforms, including AI vision gateways, fleet video telematics, surround-view, video recording, and passive driver-assistance systems. Compared with CV5, CV7 reduces power consumption by about 20 per cent and delivers more than 2.5 times higher AI throughput through a third-generation CVflow AI accelerator that supports concurrent CNN and transformer-based networks. The SoC integrates an AI accelerator, an image signal processor, video encoding, ARM